

SAMMY MUSTAFA

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RESEARCH INTERESTS

Artificial Intelligence and Machine Learning in Healthcare, Clinical Trial Informatics, Healthcare Equity and Access, Translational Biomedical Informatics, Digital Health Technologies, Precision Medicine

EDUCATION

Feinberg School of Medicine Chicago, IL
MD Incoming 2026, Expected 2030

Harvard Medical School Boston, MA
M.S., Biomedical Informatics (GPA: 3.97) 2024

Northwestern University Evanston, IL
B.A., Biological Sciences, Minors in Data Science & Linguistics (GPA 3.93) 2023

PROFESSIONAL EXPERIENCE

Bond Health New York, NY
Co-Founder & CIO Jan 2024 – Present

- Build proprietary LLM-based platform to improve the efficiency and inclusivity of clinical trial matching and patient screening
- Lead regulatory and compliance strategy (HIPAA, BAA agreements, and data security frameworks) for pilots with academic health systems and pharmaceutical sponsors
- Direct technical roadmap for ML-based patient screening and trial-matching pipelines, ensuring adaptability across diverse populations and clinical indications
- Conduct stakeholder interviews with principal investigators, clinical research coordinators, patients, and pharmaceutical sponsors to guide product design

Behavior Remote
Data Scientist Oct 2023 – Dec 2023

- Applied RNNs and ensemble models on 200,000+ multimodal physiological datapoints to predict craving episodes in substance use recovery patients
- Enhanced unsupervised learning pipelines by integrating label propagation with scikit-learn clustering, achieving a 15% improvement in predictive precision for stratification
- Optimized the Cypher algorithm for real-time anomaly detection in wearable device streams, enabling faster intervention alerts and improved patient monitoring
- Contributed to development of digital therapeutic tools combining wearable sensors, ML models, and behavioral interventions to support long-term addiction recovery

H2Ok Innovations Boston, MA
Associate Consultant Sep 2023 – Dec 2023

- Led the implementation of AI-powered, IoT-enabled systems optimizing industrial liquid management, deploying scalable solutions for Fortune 500 clients
- Served as project manager for strategic market and sensitivity analyses in a potential \$14Bn market segment, identifying key entry points and value propositions
- Boosted revenue potential by 25% by designing bespoke pricing strategies grounded in competitive benchmarking and customer needs analysis
- Collaborated with cross-functional teams (data scientists, engineers, and business development) to align technical feasibility with regulatory considerations and market

Mercurial-AI

Chicago, IL

Strategy Consultant

Mar 2023 – Aug 2023

- Spearheaded operational reconfiguration and go-to-market strategy that enabled the company to secure over \$300,000 in seed funding
- Served as project lead for developing a deep-ensemble and causal modeling framework to tailor treatment recommendations for breast cancer patients
- Conducted comprehensive competitive landscape and total addressable market (TAM) analyses, providing strategic insights for oncology product positioning and differentiation
- Prepared investor materials and strategic briefs that translated technical milestones into compelling business propositions for venture capital and clinical collaborators

Northwestern University Residential Service

Evanston, IL

Resident Assistant

Aug 2021 – Jun 2023

- Oversaw 44 residents, served overnight duty shifts, worked the front desk, and developed community builder events
- Represented the RAs on the RA Advisory Board, advocated for adjustments to the position and services

RESEARCH EXPERIENCE

Center for Precision Psychiatry, Massachusetts General Hospital

Boston, MA

ML Researcher

Dec 2023 – Dec 2024

- Developed a metabolomics framework for early detection of major depressive disorder (MDD), stratified by severity and phenotype
- Implemented multivariable logistic regression, mutual information pipelines, and Louvain clustering across 280,000 UK Biobank participants and 250+ metabolite features
- Identified phenotype-specific lipid biomarkers predictive of MDD heterogeneity, informing personalized psychiatry approaches
- Applied ML-based residualization models to separate genetic vs. environmental contributions to metabolic biomarkers

Potocsnak Longevity Institute, Feinberg School of Medicine

Chicago, IL

Molecular Biology Researcher

Mar 2021 – Jun 2023

- Probed the RNA-binding protein SERBP1 as a regulator of PAI-1, a central mediator of tissue fibrosis, senescence, aging, and Alzheimer's disease
- Designed and executed an independent project using GFP-tagged SERBP1 to study subcellular localization under high glucose and small-molecule inhibitor treatments

- Identified SERBP1's potential prion-like, transgenerational behavior with implications for epigenetic regulation of PAI-1 expression and long-term fibrosis risk
- Proposed therapeutic strategies targeting SERBP1 localization and binding to destabilize PAI-1 mRNA for potential cardiovascular and aging-related disease intervention

Behavioral Research Lab, Yale University

Remote

NLP Research Associate

Mar 2022 – Jul 2022

- Conducted comparative analyses of bargaining behaviors in verbal vs. non-verbal negotiation experiments, supporting a large-scale behavioral economics + NLP dataset
- Analyzed over 6,000 minutes of negotiation transcripts using Python using sentimental analyses, word embeddings, etc.) to quantify and classify linguistic behaviors
- Assisted in annotating thousands of speech acts (pushes, comparisons, allowances, offers), helping establish one of the largest negotiation dialogue corpora to date
- Findings revealed that verbal communication led to faster convergence, higher agreement rates, and reduced variance in negotiated prices

Complex Systems Lab, Polytechnic Institute of Catalonia

Barcelona, Spain

Systems Biology Research Assistant

Jun 2022 – Sep 2022

- Designed mathematical and spatial resolution models of the cAMP-PDE signaling pathway in atrial myocytes via MATLAB
- Developed differential formulas to model the autonomic effects of atrial fibrillation risk variant rs13143308T
- Investigated the modulation of the PDE1-Adenosine A_{2A} macromolecular cluster on sarcoplasmic/cytosolic calcium levels

PEER-REVIEWED PUBLICATIONS

- **Mustafa S**, Moraitaki H, Celi LA, Hao S, Goode D, Kahan T. Evaluating pain assessment and management in acute pancreatitis. *MIT Health and Life Sciences (HEALS) Conference*, Cambridge, MA, December 4, 2024. Preprint: <https://sammymustafa.com/research/heals.pdf>
- Sharon DA, Huang Y, Oyewole M, **Mustafa S**. How to Go With the Flow: An Analysis of Flow Matching Molecular Docking Performance With Priors of Varying Information Content. *ICLR 2024 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, Vienna, Austria, May 11, 2024. OpenReview: <https://openreview.net/forum?id=FGwGt9XWO4>

MANUSCRIPTS UNDER REVIEW

- **Mustafa S**, Shachar C. A Medico-Legal Approach to Subjective Pain Testimony in Disability Adjudication. Harvard Law School. 2024. Paper: <https://sammymustafa.com/research/hls.pdf>
- Chen Y, **Mustafa S**, Picard R. Integrating Anticipatory and Physiological Factors to Predict Injection Pain Intensity. Affective Computing Lab, MIT Media Lab. 2024. Paper: <https://sammymustafa.com/research/media.pdf>

- **Mustafa S**, Watts D, Smoller JW. Metabolic Biomarkers of MDD Phenotype Heterogeneity. Center for Precision Psychiatry, Massachusetts General Hospital. 2024. Paper: <https://sammymustafa.com/research/psych.pdf>
- Ameen S, Chen X, Liu G, **Mustafa S**, Rigby T. Diagnosing Autism Spectrum Disorder from 3D MRI Images in Pediatric Cases. Department of Biomedical Informatics, Harvard Medical School. 2024. Paper: <https://sammymustafa.com/research/mri.pdf>
- **Mustafa S**, Vaughan D. SERBP1: Exploiting RNA-binding protein-mediated PAI-1 inhibition. Potocsnak Longevity Institute, Northwestern University Feinberg School of Medicine. 2022. Paper: <https://sammymustafa.com/research/serbp1.pdf>
- **Mustafa S**, Leonardi D. PTENP1: A Pivotal Pseudogene in Glioblastomas. Cell Biology Lab, Bergen County Academies. 2020. Paper: <https://sammymustafa.com/research/ptenp1.pdf>

HONORS/AWARDS/FUNDING

\$150K CEAS SAFE Investment at \$3M Valuation	2025
Harvard Law Entrepreneurship Project Awardee	2025
NSF I-Corps Spark/Fusion Program Grant Awardee (\$1,500)	2025
MIT Sandbox Innovation Fund Awardee (\$2,500)	2025
MIT 100K Pitch Competition Semifinalist (\$2,000)	2024
Top 20 Team @ Penn Healthcare Case Competition	2023
Weinberg Research Grant (\$4,000)	2022
Honors Program in Medical Education Research Grant (\$3,000)	2021
Thermo Fisher Scientific Antibody Scholarship Award (\$5,000)	2020
Regeneron Science Talent Search Scholar (\$2,000)	2020

SERVICE & COMMUNITY OUTREACH

Cambridge After School Program (CASP)	Cambridge, MA
<i>After-School Volunteer</i>	<i>Jun 2024 – Dec 2024</i>
• Provided homework help, mentorship, and enrichment activities for low-income, immigrant youth in Cambridge (grades 3-8)	
Lifestyle Interventions for Youth (LIVELY)	Boston, MA
<i>Workshop Facilitator</i>	<i>May 2024 – Dec 2024</i>
• Led interactive workshops on lifestyle medicine pillars (nutrition, physical activity, sleep, stress management, substance avoidance, and social connections) for youth ages 13-14	

- Promoted early health education by leading peer discussions that connected physical and mental health, empowering adolescents at several Boston Boys & Girls Clubs

Camp Kesem

Evanston, IL

Unit Leader

Sep 2022 – Aug 2023

- Served as Unit Leader at two 5-day camps for children (ages 10-12) whose parents/guardians were affected by cancer, ensuring well-being, safety, and inclusion in activities
- Oversaw medication schedules, daily routines, and emotional support for ~20 campers, fostering a supportive community environment
- Facilitated group discussions and recreational activities that encouraged peer connection, empowerment, and coping strategies for youth impacted by parental cancer

Books & Breakfast (B&B)

Evanston, IL

Before-School Tutor & Mentor

Sep 2022 – Aug 2023

- Tutored and mentored elementary students from low-income and immigrant backgrounds to improve academic readiness, literacy, and confidence before the school day
- Supported an equity-focused program providing nutritious breakfasts, 1-1 English tutoring in the summer, and holistic support

Evanston Grows

Evanston, IL

Community Gardener Volunteer

Sep 2022 – Jun 2023

- Supported community-led initiatives to reduce food insecurity by growing local produce and strengthening neighborhood food distribution networks

PROFESSIONAL ASSOCIATIONS

American Association for the Advancement of Science (AAAS)	<i>2025 – Present</i>
NIH CTSA Recruitment Innovation Center (RIC) Community of Practice	<i>2025 – Present</i>
Illinois Alliance to Promote Opportunities for Health (IAPOH)	<i>2025 – Present</i>
Institute of Electrical and Electronics Engineers	<i>2024 – Present</i>